

Experiment 4.3

Determination and plotting of Electric Field Intensity due to Charged Disk

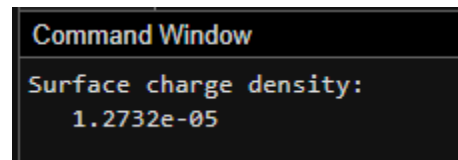
Code File Name: Exp4_3_ChargedDisk_EField.m

Code

```
clc;
R = 0.5;
Q = 10e-6;
A = pi*R^2;
sigma = Q/A;
disp("Surface charge density:");
disp(sigma);
z = linspace(0.1,2,100);
k = 9e9;
E = (k*sigma*R^2)./(2*(z.^2 + R^2).^(3/2));
plot(z,E,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Electric Field (N/C)');
title('Electric Field due to Charged Disk');
```

Command Prompt

Surface charge density:
1.2732e-05



Output Graph

